

8. Reflection What is useful about implementing a neural network manually? Implementing a neural network manually using NumPy removes the "black box" nature of modern frameworks like PyTorch or TensorFlow. It requires a deep, fundamental understanding of linear algebra (matrix dot products for weight mapping) and calculus (the chain rule required for backpropagation). Manually coding the parameter updates forces an understanding of exactly how learning rates and gradients interact to shift the network's internal logic.

What are the limitations of using MSE for a binary classification task? Using Mean Squared Error (MSE) combined with a Sigmoid activation is fundamentally flawed for binary classification. MSE measures raw distance, which is ideal for predicting continuous numbers (like housing prices), but binary classification relies on probabilities. As predictions approach the extremes (0 or 1), the derivative of the Sigmoid function becomes incredibly close to zero. This leads to the "vanishing gradient" problem, where the network gets stuck and stops updating its weights, which is exactly why our model struggled to find a strong decision boundary. Modern binary classifiers use Binary Cross-Entropy (Log Loss) instead, which mathematically corrects this issue.

What information is missing from this admissions dataset? This dataset is missing the qualitative, holistic data that drives actual graduate admissions. It lacks the applicant's Statement of Purpose, Letters of Recommendation, undergraduate research experience, relevant publications, interview scores, and the prestige of their undergraduate institution.

Why might this model produce misleading results even if its test accuracy looks reasonable? A model can achieve an artificially "reasonable" accuracy simply by guessing the majority class. If 60% of applicants in a dataset are rejected, a model that blindly guesses "Rejected" every single time will achieve 60% accuracy without actually learning anything about the features. Furthermore, the model might learn systemic biases in the data (such as heavily penalizing international applicants due to real-world visa/funding bottlenecks) rather than evaluating true academic merit, leading to technically "accurate" but highly misleading or biased predictions.

What would you change if you wanted to make this model stronger or more realistic? To make the model stronger mathematically, I would change the loss function from Mean Squared Error to Binary Cross-Entropy. To make the model more realistic, I would require a much richer dataset. Incorporating Natural Language Processing (NLP) sentiment scores derived from recommendation letters or application essays would give the model visibility into the holistic traits that actual admissions committees use to differentiate candidates with identical GPAs.